生物统计和生物信息引论

Lecture 13

Large-scale multiple testing

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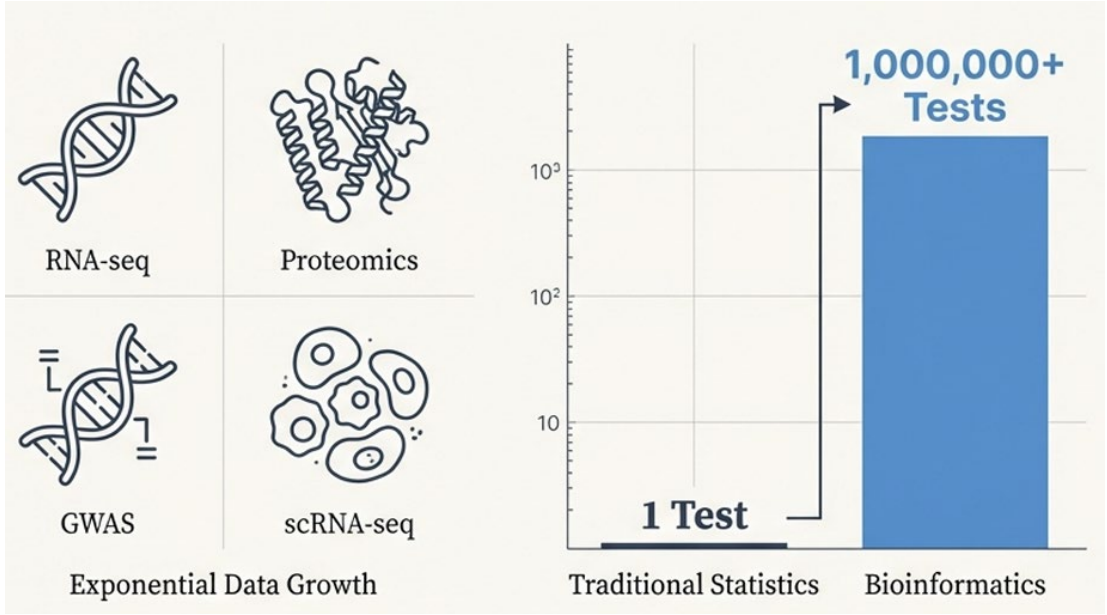
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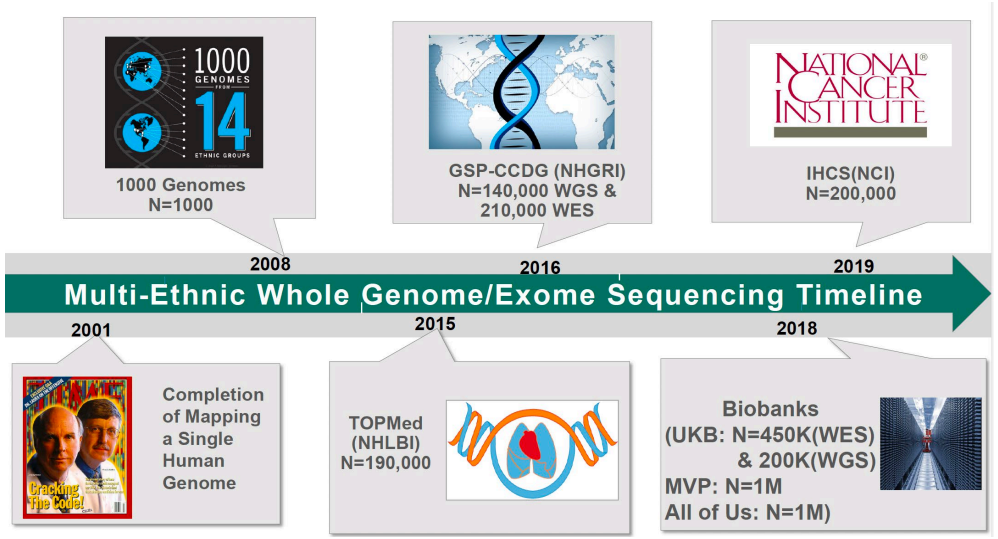
Our Data is Exploding.
So Are Our Chances to be Wrong

High-throughput data generates thousands/millions of simultaneous hypotheses.

How to define the significance & find interesting discovery?

Reproducibility? Reliable biological insights





Hypothesis Testing and P-value

Hypothesis Testing

In regression $Y = \beta_0 + \beta_1 X_1 + \epsilon$, we test:

- **Null (H_0):** $\beta_1 = 0$ (No effect)
- **Alternative (H_1):** $\beta_1 \neq 0$ (Effect exists)

```
             Estimate   Std. Error  t value   Pr(>|t|)
(Intercept)   5.79778    2.76324    2.098    0.0385 *
x             0.47376    0.05344    8.865   3.5e-14 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1
```

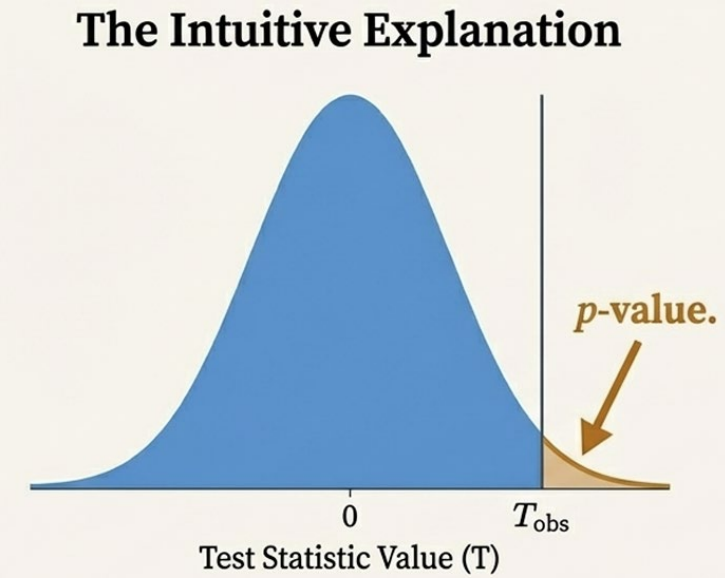
It quantifies the evidence against the null hypothesis.

Let T be a test statistic, then the p-value is:

$$p = P(T \geq T_{obs} | H_0)$$

where T_{obs} is the observed value of the test statistic.

The probability of observing a result at least as extreme as the data under



Null Hypothesis and Interpretation

Wilcoxon test

- Non-parametric: Compare two distribution difference
- Mean difference / the dispersion difference

GLM test

- Parametric: coefficient test
- Model the mean difference

Classical Interpretation

α	0.1	0.05	0.025	0.01
Strength of Evidence	Borderline	Moderate	Substantial	Strong

- With big sample sizes, it is easy to have small p-value even when the estimated effect is small.
- performing many tests can lead to many false discoveries
- It does not measure reproducibility

Large-scale inference (Finding the needle in a haystack)

Test n hypotheses $H_{01}, H_{02}, \cdot \cdot , H_{0n}$ with a collection of p-values $P_1, P_2, \cdot \cdot , P_n$

1. Global Detection

"Is there any needle?"

Testing the global null hypothesis $H_0 = \cap H_{0i}$. We want to know if *any* signal exists in the entire dataset.

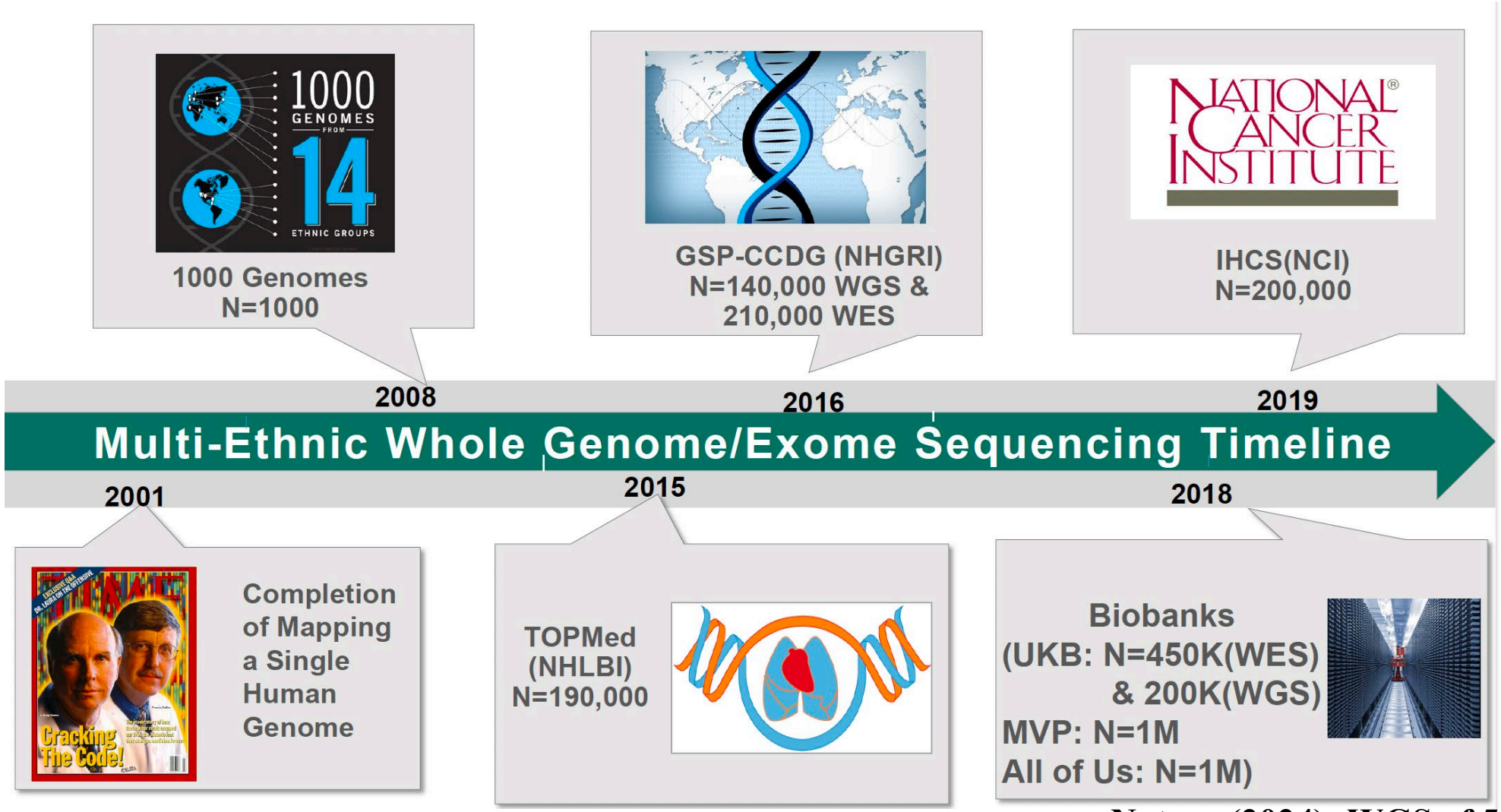


2. Identification

"Where is the needle?"

Identifying exactly *which* individual hypotheses are false. This is a classification problem balancing discovery and error.

现代群体遗传数据



Nature (2024): WGS of 500,000 UK Biobank participants

更多

Mexican Biobank (N = 140K)

- Cohort (Nature 2023)
- Association+ Genetic prediction (Nature 2023)

Taiwan Precision Medicine Initiative (N = 500K)

- Cohort (Nature 2025)
- Association + Genetic prediction (Nature 2025)

新闻资讯｜未来十年，英格兰将对所有新生儿进行全基因组测序

ChatGPT Human Genetics Network 2025年6月23日 17:32 北京 11人

英国政府近日宣布了一项重磅计划：未来十年，英格兰所有新生儿将在出生后进行全基因组测序，以评估其患上数百种疾病的风险。这项举措是NHS十年战略的一部分，旨在将医疗服务从“诊断与治疗”转型为“预测与预防”，以缓解日益加重的医疗资源压力。该计划将投入6.5亿英镑用于DNA研究和相关技术发展，并计划到2030年完成50万份基因组数据的积累，打造全球最大的人类基因数据库之一。

Genomic Organization: From Chromosome to SNP

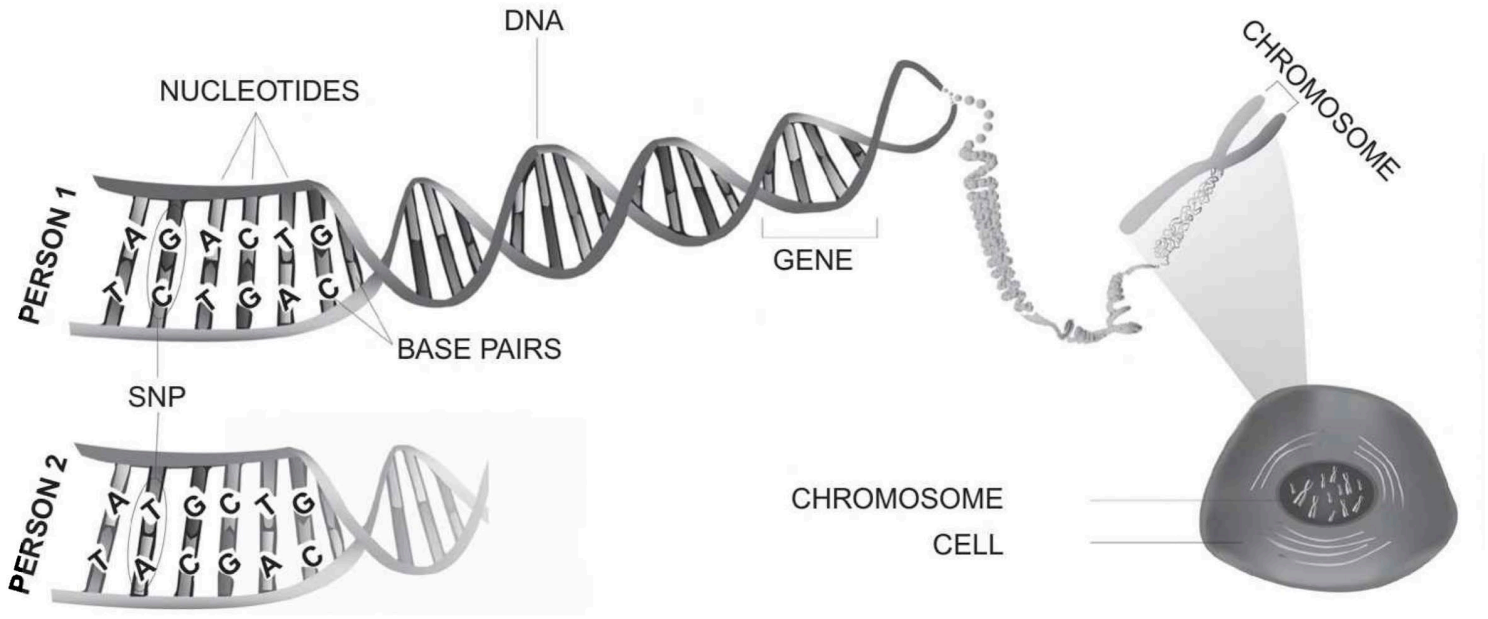


Figure 1.1
Orgaization of DNA (deoxyribonucleic acid) in the cell nucleus.

Chromosome
Chromosome = DNA packaged with proteins.

DNA
DNA is the linear nucleotide sequence inside that chromosome.

Gene
A gene is one defined functional segment along the DNA sequence.

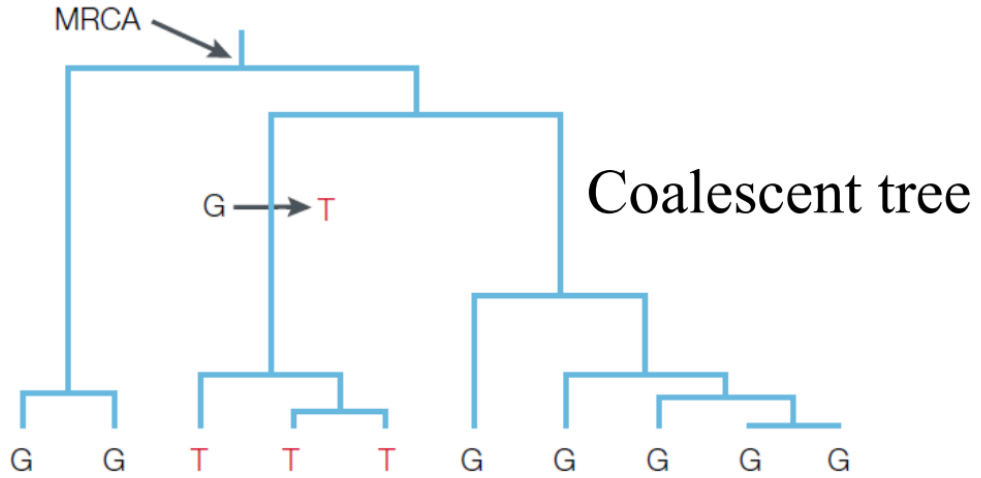
SNP
A SNP (single nucleotide polymorphism) is a single-base variant at one genomic position.

What is a SNP?

A **S**ingle **N**ucleotide **P**olymorphism (**SNP**) is a letter of the genome that differs in different individuals (e.g. G/T).

Each SNP corresponds to one single mutation event in history, e.g. G mutated to T in one single ancestor.
G = **ancestral** allele, T = **derived** allele.





Coalescent tree

Rosenberg & Nordborg 2002 Nat Rev Genet

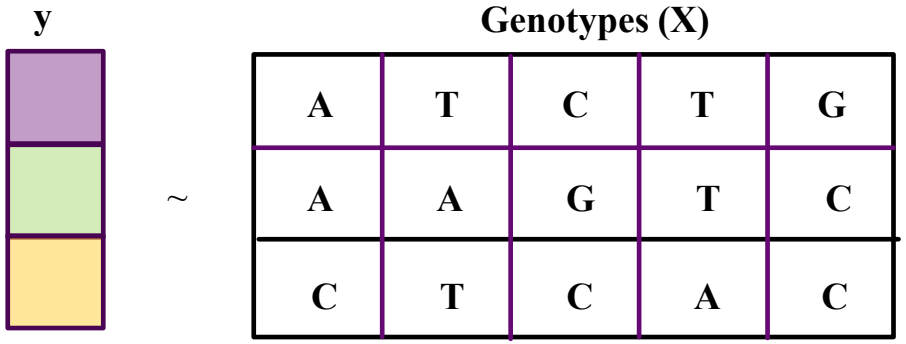
Massive Rare Variants in WGS Data (~97% Non-coding) (TOPMed Freeze 10a (n=185,000))

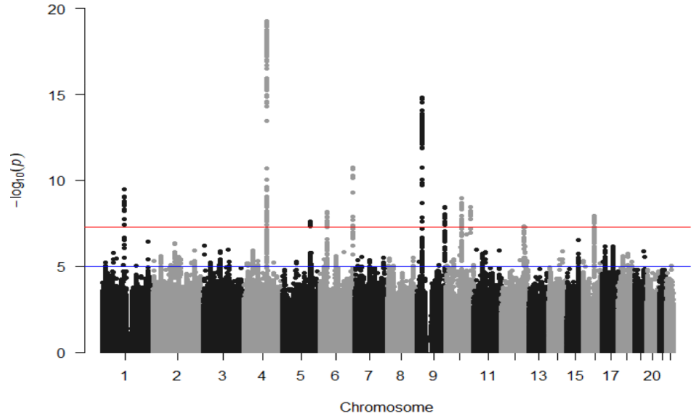
Type	Category	PASS Variants	Singletons	Doubletons	MAF > .0001	MAF > .001	MAF > .005	MAF > .05
SNP	All	845M	44.7%	15.6%	4.61%	1.18%	0.98%	0.75%
	Synonymous	3.17M	41.0%	15.1%	5.29%	1.28%	0.97%	0.66%
	Missense	6.74M	45.4%	15.7%	3.91%	0.79%	0.51%	0.28%
	pLoF	442K	52.2%	16.2%	2.53%	0.52%	0.35%	0.23%
Indels	All	70.5M	46.6%	15.5%	4.53%	1.07%	0.77%	0.46%
	Inframe	91K	32.9%	15.8%	6.07%	0.98%	0.43%	0.17%
	Frameshift	182K	43.7%	17.5%	2.94%	0.50%	0.27%	0.12%

Pre-QC: 1 billion+ variants

研究背景和目标

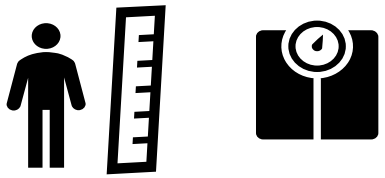
揭示各种与复杂疾病和性状相关的遗传变异



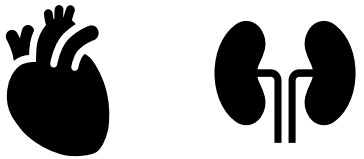


不同性状/疾病+共同的遗传关联位点 ⇨ 遗传结构/遗传相关性

(shared genetic architecture)



身高 & 体重



心血管疾病
& 肾功能

探测(detection) **识别**(identification) **刻画** (quantification) **估计** (estimation)

Biomedical Motivation: GWAS

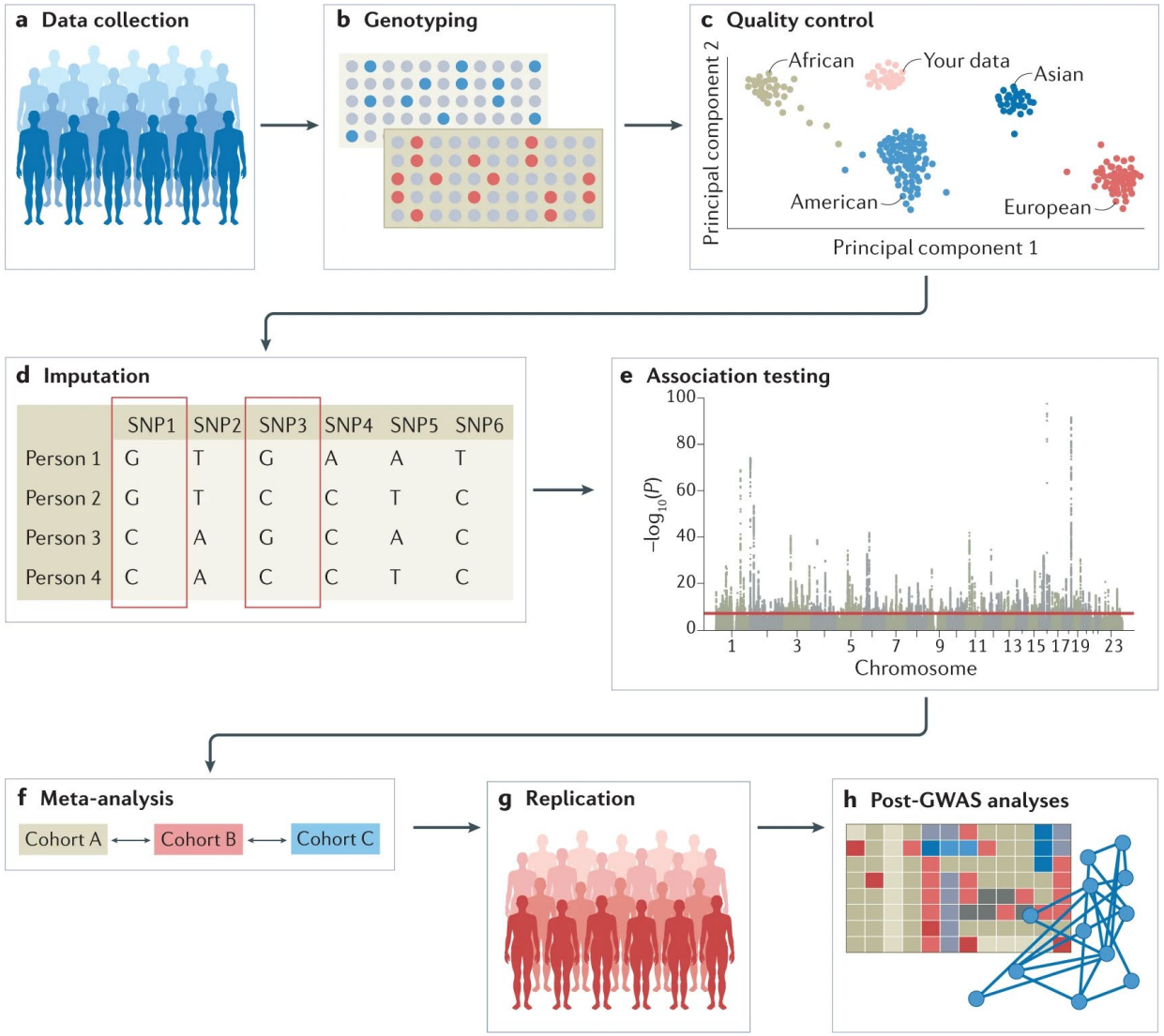
A GWAS studies whether genetic variants are associated with a disease or trait.

For example, suppose we study a disease such as type 2 diabetes; in this file, $G = 2,473,441$ SNPs are tested.

Main columns include SNP identifier and association text p-value.

	SNP str	CHROMOSOME int	POSITION int	RISK_ALLELE str	OTHER_ALLELE str	P_VALUE dbl
1	rs12565286	1	711153	G	C	0.1600
2	rs3094315	1	742429	A	G	0.8200
3	rs2905035	1	765522	G	A	0.3600
4	rs2980319	1	766985	T	A	0.3900
5	rs4040617	1	769185	A	G	0.5000
6	rs2977612	1	770648	A	T	0.3500
7	rs2905062	1	774913	A	G	0.3300
8	rs2980300	1	775852	C	T	0.6100
9	rs11240777	1	788822	G	A	0.7800
10	rs17160906	1	793404	G	A	0.3400
11	rs3121561	1	980243	C	T	0.3000
12	rs3813193	1	988364	G	C	0.6600
13	rs4075116	1	993492	T	C	0.9700
14	rs3934834	1	995669	C	T	0.4300
15	rs3766193	1	1007033	G	C	0.2900
16	rs3766192	1	1007060	T	C	0.4700
17	rs3766191	1	1007450	C	T	0.4100
18	rs9442371	1	1008425	T	C	0.4800
19	rs9442372	1	1008567	G	A	0.6900
20	rs10907177	1	1011209	A	G	0.4200
21	rs3737728	1	1011278	G	A	0.7300
22	rs10907178	1	1011446	A	C	0.4300

Genome-wide Association Studies: The Full Pipeline



GWAS

A genome-wide association study (GWAS) is a research approach that scans the genomes of large populations to identify genetic variants (usually SNPs) statistically associated with specific diseases or traits.

Source: Uffelmann E et al. Genome-wide association studies. Nat Rev Methods Primers. 2021. doi:10.1038/s43586-021-00056-9

GWAS question

The scientific question: Which SNPs are associates with disease risk?

For SNP $g = 1, \ldots, G$, we test:

$$H_{0g}: \text{ SNP } g \text{ is not associated with disease}$$

against

$$H_{1g}: \text{ SNP } g \text{ is associated with disease.}$$

Each SNP gives one p-value: $p_1, p_2, \ldots, p_G$.

Data

X is 0/1/2.

For example, suppose we have a biallelic SNP whose reference allele is A and the alternative allele is G.

There are three possible genotypes for this SNP: AA, AG, and GG.

Additive model is most common.

A study usually contains 1e6~1e7 SNPs.

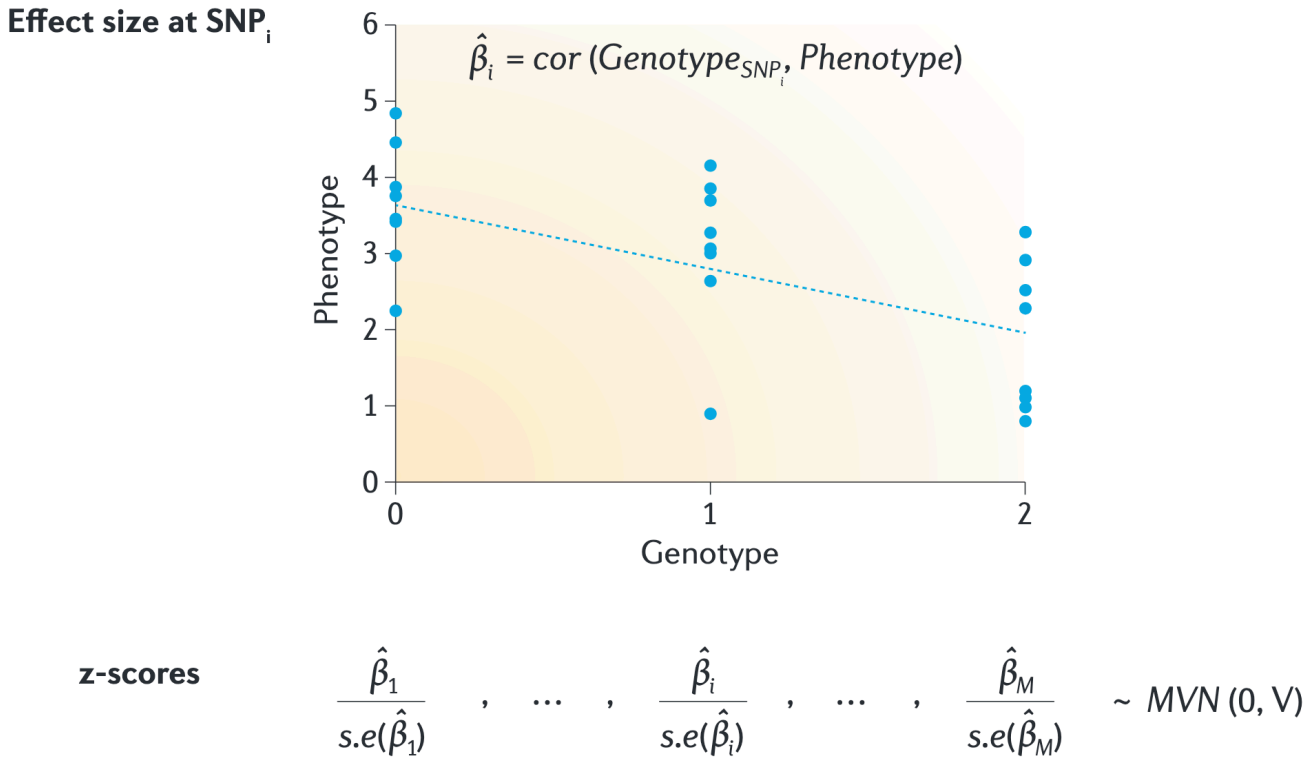
Genetic models	AA	AG	GG
Additive model	0	1	2
Dominant model	0	1	1
Recessive model	0	0	1

Y is outcome.

Can be blood glucose/heart failure risk/…

Y can be binary (e.g. Type 2 diabetes or not)

Summary statistics compress each SNP into an interpretable estimate



Implement univariate regression for each SNP

$$y = x_i\beta_i + \epsilon, \quad \epsilon \sim N(0, \sigma^2)$$

$$\hat{\beta} = \frac{x^T y}{x^T x} \qquad x, y \text{ standardized}$$

$$z_i = \frac{\hat{\beta}_i}{\mathrm{se}(\hat{\beta}_i)}$$

$$p_i = 2\Phi(-|z_i|)$$

Source: Pasaniuc B, Price AL. Dissecting the genetics of complex traits using summary association statistics. Nat Rev Genet. 2017.

Why classical testing fails

For one single test, significance level 0.05 means:

$$\Pr(\text{false positive}) = 0.05$$

if the null hypothesis is true.

But if we test $G = 2,473,441$ SNPs at level 0.05, then even if all null hypotheses are true, the expected number of false positives is:

$$G \times 0.05 = 2,473,441 \times 0.05 \approx 123,672.$$

Our core question: How can we find true disease-associated SNPs while controlling false positives?

P-value histogram intuition

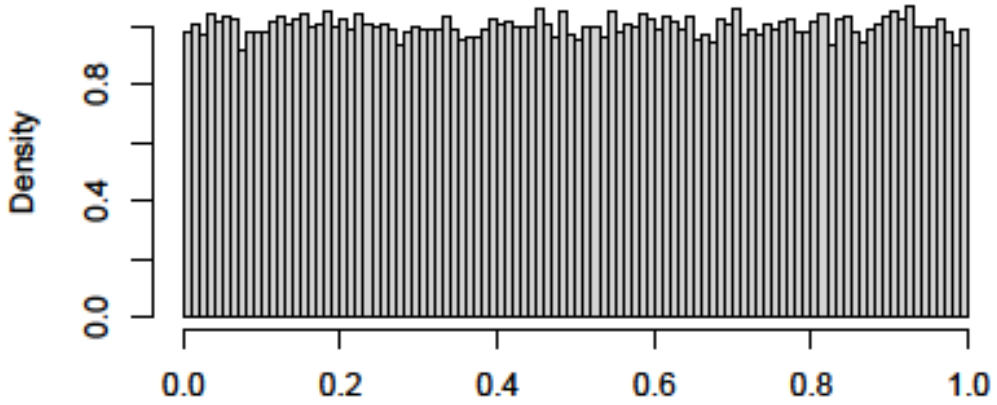
Under a valid null hypothesis, $p_g \sim \text{Uniform}(0, 1)$.

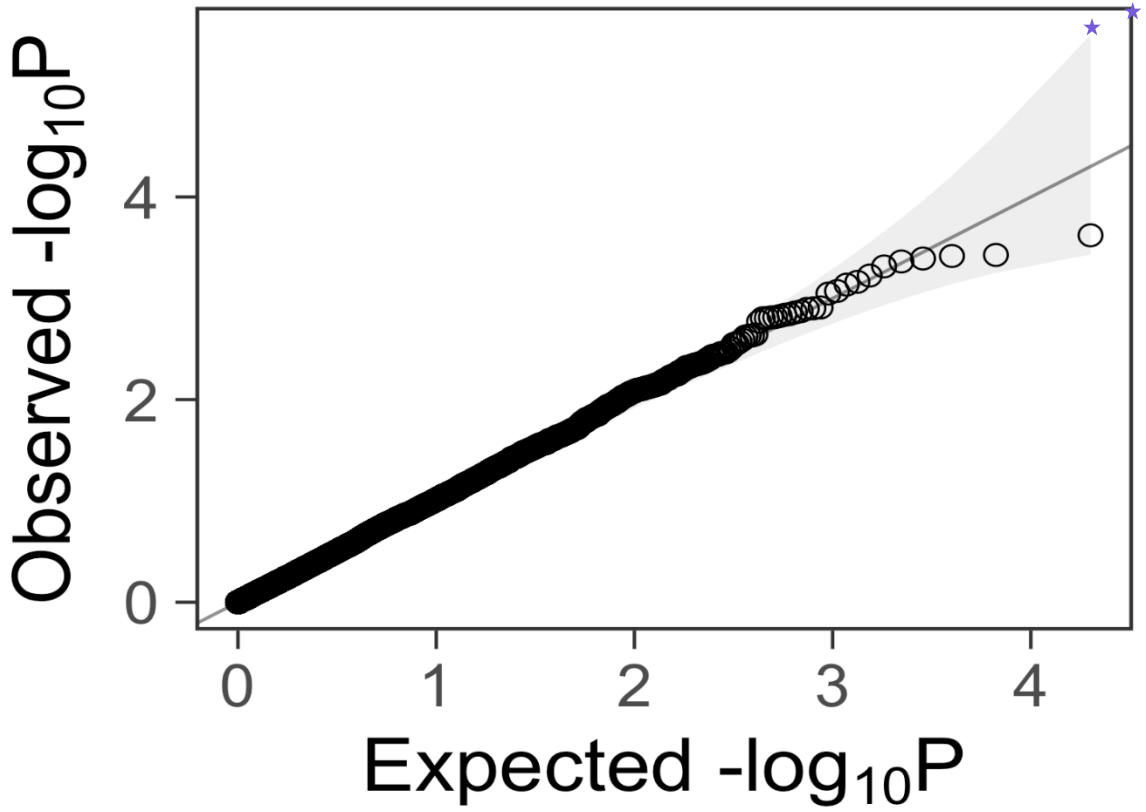
If most SNPs are not associated with disease, then most p-values should look approximately uniform.

If some SNPs are truly associated with disease, their p-values tend to be very small.

Quality control of testing statistics

Understand your data: distribution of p-values



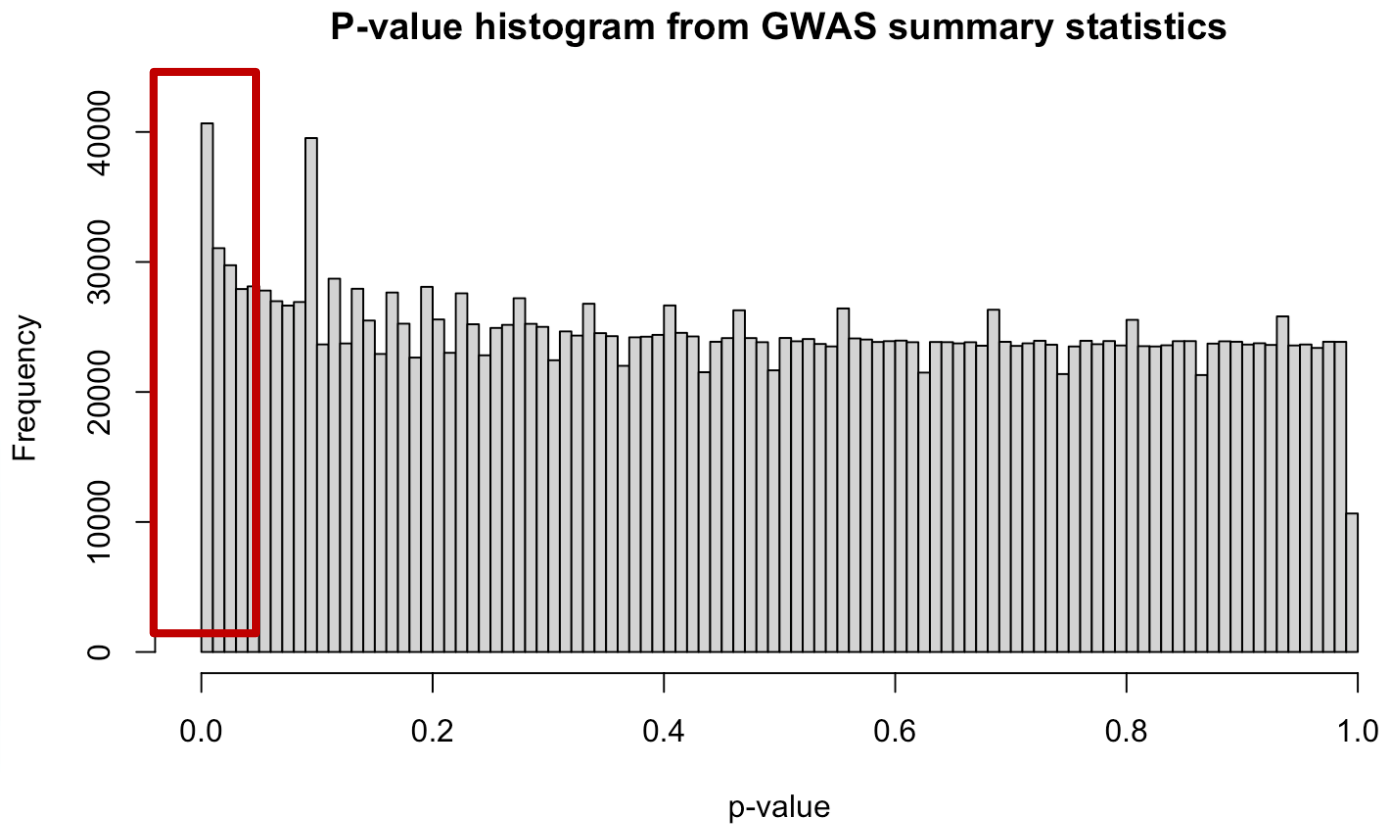


P-value histogram


```r
dat ← read.delim(
  "DIAGRAMv3.2012DEC17.txt",
  header = TRUE,
  stringsAsFactors = FALSE
)

library(ggplot2)

hist(dat$P_VALUE, breaks = 80, main = "P-value
histogram from GWAS summary statistics", xlab =
"p-value")
```



Why GWAS needs a stronger threshold

In our dataset, there are $G = 2,473,441$ SNP-level tests. A Bonferroni threshold would be:

$$\frac{0.05}{G} = \frac{0.05}{2,473,441} \approx 2.02 \times 10^{-8}.$$

In GWAS, a common significance threshold is: 5×10^{-8}, which is **much smaller** than 0.05. This is because many nearby SNPs carry similar genetic information because of linkage disequilibrium (LD); therefore, the effective number of independent tests is much smaller than the total number of measured SNPs.

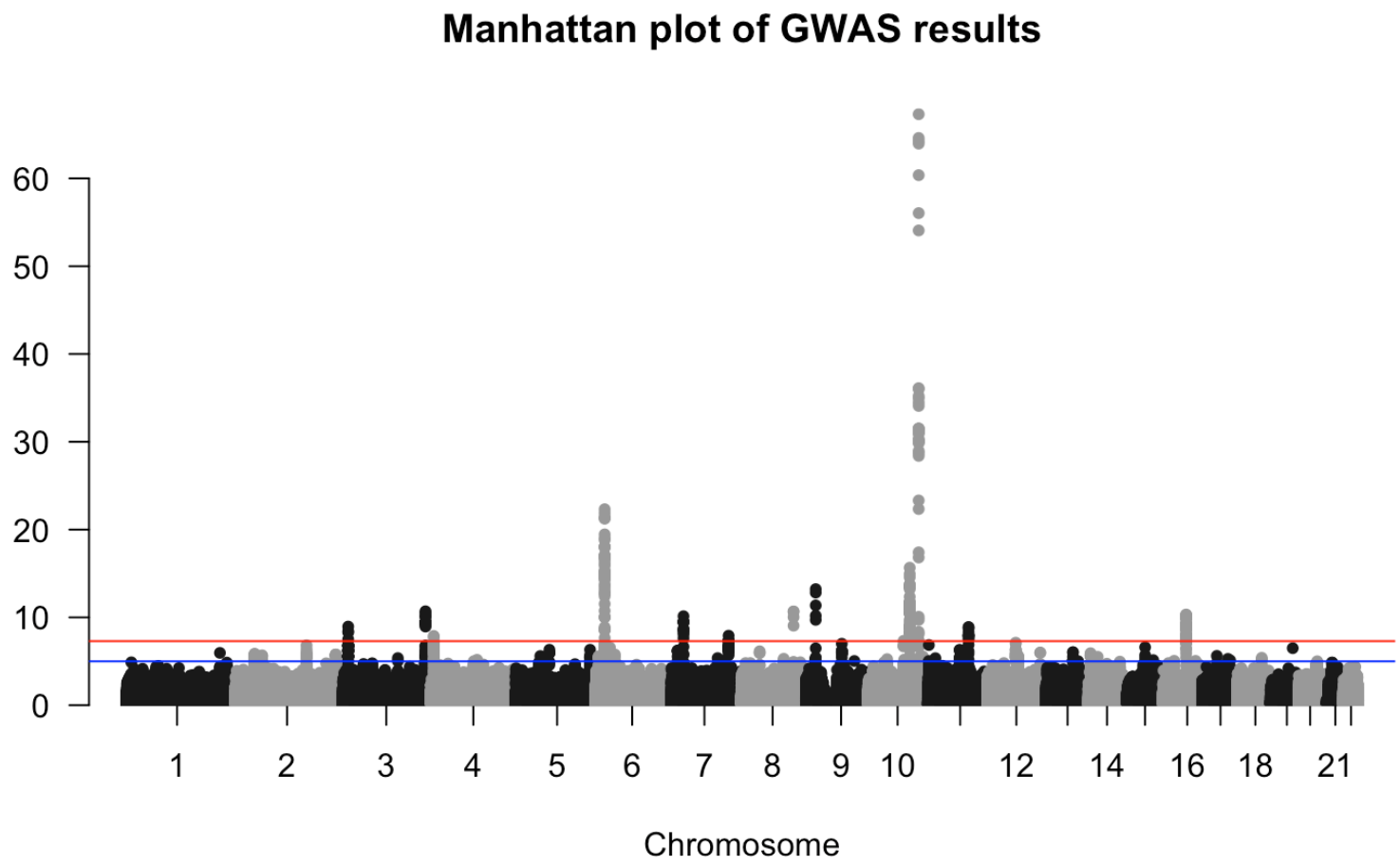
A commonly used approximation is about 10^6 independent tests, which leads to the threshold $0.05/10^6 = 5 \times 10^{-8}$.

Manhattan plot

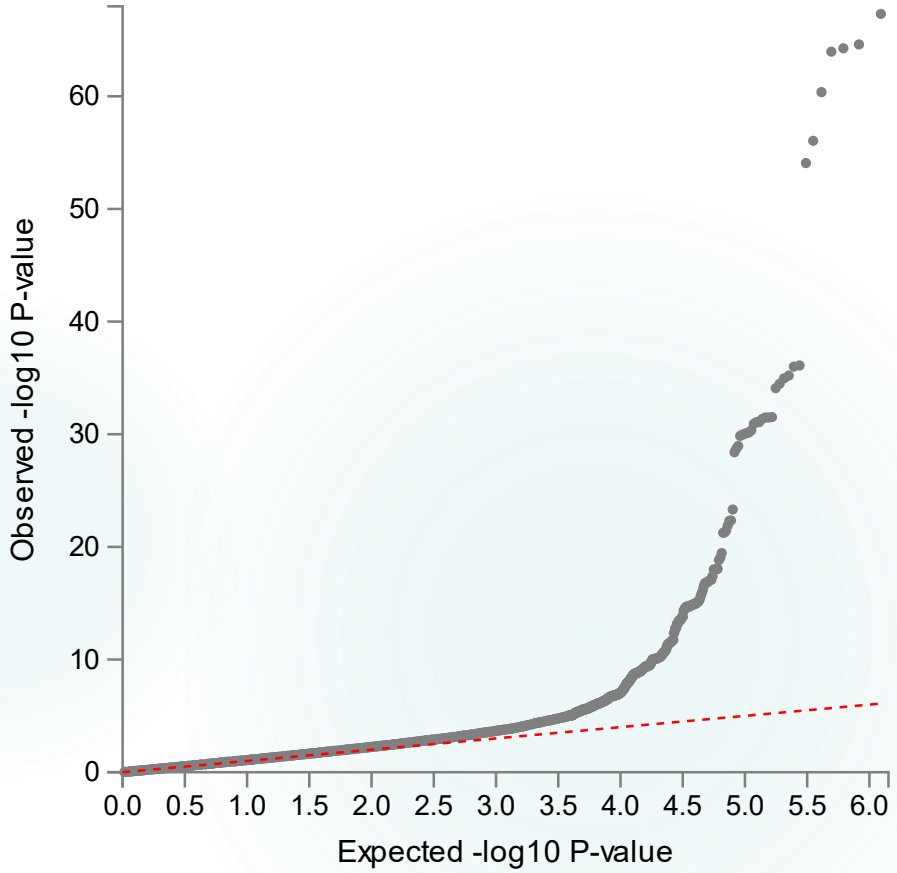

```r
library(qqman)

gwas_for_plot ← data.frame(
  SNP = dat$SNP,
  CHR = dat$CHROMOSOME,
  BP = dat$POSITION,
  P = dat$P_VALUE
)

manhattan(gwas_for_plot,
          chr = "CHR",
          bp = "BP",
          snp = "SNP",
          p = "P",
          genomewideline = -log10(5e-8),
          suggestiveline = -log10(1e-5),
          main = "Manhattan plot of GWAS
          results")
```

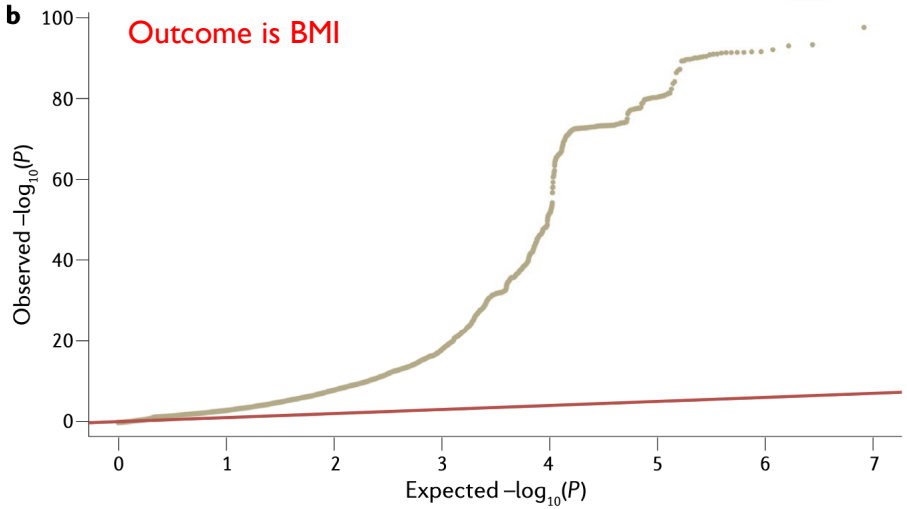


QQ plot



We usually assume only a small number of SNPs are associated with the disease.
Under null hypothesis, p-value should be uniformly distributed between 0 and 1.

Deviation from the expectation under the null hypothesis (red line) indicates the presence of either true causal effects or insufficiently corrected population stratification. E.g. the BMI QQ-Plot



Top SNPs in the GWAS data

In the GWAS data, the smallest p-values can be extremely small. For example, the top SNPs include signals on chromosome 10 with p-values much smaller than 5×10^{-8}.

SNP str	CHROMOSOME int	POSITION int	RISK_ALLELE str	OTHER_ALLELE str	P_VALUE dbl
rs4506565	10	114746031	T	A	4.90e-68
rs7901695	10	114744078	C	T	2.50e-65
rs7903146	10	114748339	T	C	5.50e-65
rs4132670	10	114757761	A	G	1.10e-64
rs12243326	10	114778805	C	T	4.30e-61
rs12255372	10	114798892	T	G	9.00e-57
rs17747324	10	114742493	C	T	8.50e-55
rs4074720	10	114738487	T	C	7.80e-37
rs7077039	10	114779067	C	T	1.00e-36
rs6585201	10	114758773	A	G	6.20e-36
rs10787472	10	114771287	C	A	1.10e-35
rs11196208	10	114801306	C	T	3.30e-35
rs11196205	10	114797037	C	G	8.20e-35
rs11196200	10	114791928	G	C	3.00e-32
rs10885402	10	114751687	C	A	3.30e-32
rs10885409	10	114798062	C	T	3.30e-32
rs4074718	10	114738607	A	G	4.50e-32
rs7100927	10	114786038	G	A	8.40e-32
rs7895340	10	114791515	A	G	8.40e-32
rs7900150	10	114783813	A	T	1.20e-31
rs6585200	10	114758599	G	A	4.60e-31
rs10885405	10	114767660	T	C	6.90e-31
rs6585198	10	114752227	A	G	7.80e-31

Error table for GWAS discoveries

Reality / Decision	**Not significant**	**Significant**
Null SNP	U	V
Associated SNP	T	S

- V: false positive SNPs
- S: true positive SNPs

Main target: many true discoveries S, but few false discoveries V.

FWER in GWAS

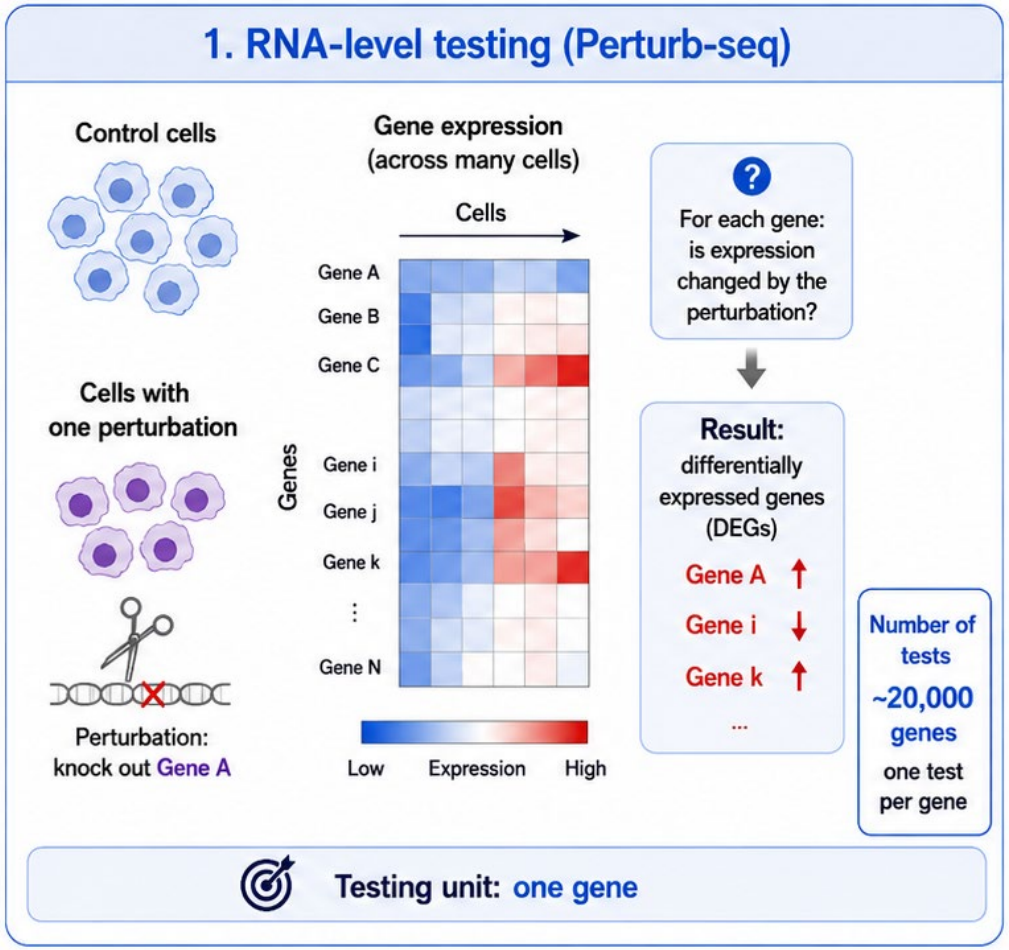
Definition:

$$\text{Family-wise error rate (FWER)} = \Pr(V \geq 1),$$

which controls the probability of making at least one false positive.
Bonferroni correction: reject SNP g if $p_g \leq \alpha/G$.
For this GWAS set, $G = 2,473,441$, so $0.05/G \approx 2.02 \times 10^{-8}$.

A commonly used GWAS threshold is $p < 5 \times 10^{-8}$. This can also be understood as a Bonferroni-type correction for testing 1M genetic variants.

RNA-seq DE test



- Compare **control cells** vs. **perturbed cells**
- For each gene, ask:
  Does expression change after perturbation?
- Testing unit: **one gene**
- Number of tests: **~20,000 genes**
- Simple methods:
  - **two-sample t-test per gene**
  - **Wilcoxon rank-sum test**
- Output: **differentially expressed genes (DEGs)**

Global Detection Methods

Bonferroni Global Test

Rejects global null if:

$$\min(P_i) \leq \frac{\alpha}{n}$$

Best for: Sparse alternatives (Finding a single strong signal).

Valid for dependent p-values.

Example: For n= 1000, reject if pmin < 0.00005.

Fisher's Combination Test

Aggregates signals using:

$$T = -2 \sum \ln(P_i)$$

Best for: Dense alternatives (Many weak signals).

Requires independent p-values (Limitation).

Combine p-values from independent studies

$$T \sim X^2(n)$$

Combination of dependent P-values

When tests are correlated (common in spatial or genomic data), methods robust to dependence are required.

Harmonic Mean P-value (HMP)

Test statistic uses the mean of the reciprocals of p-values:

$$T_{HMP} = \frac{1}{n} \sum_{i=1}^{n} \frac{1}{P_i}$$

R package: *harmonicmeanp* (Wilson, 2019).

Cauchy Combination Test (CCT)

Based on a Cauchy distribution approximation, robust under general dependence:

$$C = \sum_{i=1}^{n} w_i \tan\left(\pi\left(0.5 - P_i\right)\right)$$

R package: *ACAT* (Liu & Xie, 2019).

Why do we need FDR?

- In Perturb-seq, we do **one test per gene**
- With ~20,000 genes, this means **~20,000 p-values**
- If we use raw $p < 0.05$:
  - even under no real effect,
  - we expect about $20{,}000 \times 0.05 = 1{,}000$ false positives
- Therefore, raw p-values are not enough
- We need **multiple-testing correction**
- For DE analysis, the common choice is **FDR control**

	A	B	H	I	J	K
1	gene	perturbati...	log2fc_approx	t_stat	p_value	q_value
2	STAT1	STAT1	-0.5742889823056757	-222.86...	0.0	0.0
3	SCRN1	STAT1	-0.384951906109602...	-53.046...	0.0	0.0
4	SPATC1L	STAT1	-0.380416035657471	-46.192...	0.0	0.0
5	HELB	STAT1	-0.233808908999793...	-77.552...	0.0	0.0
6	PLCXD1	STAT1	-0.3269277544265677	-40.319...	1.6675755117761137e-278	6.029953050582426e-275
7	COL6A2	STAT1	-0.229642930252280...	-25.237...	1.2210505172661578e-126	3.679432225362022e-123
8	TNNI3	STAT1	-0.1281813139213083	-22.515...	2.369878157242625e-103	6.121056726135237e-100
9	DACT1	STAT1	-0.160408497931549...	-21.580...	7.3355235607276534e-96	1.6578283247244496e-92
10	COL6A1	STAT1	-0.158316651451273...	-16.874...	7.926242110356884e-61	1.5922939706139163e-57

A real case:
DE test between control cells and STAT1 gene knock out in H1 cell.
18080 genes in total.

False discovery rate (FDR) control

Definition:

$$\text{False Discovery Rate (FDR)} = \mathbb{E}\left(\frac{V}{V+S}\right),$$

where V is false positive discoveries and S is the true positive discoveries.

FDR controls the **expectation proportion** of false discoveries among all reported discoveries.

If FDR is controlled at level $q = 0.05$, then among the reported discoveries, the expected false discovery proportion is at most 5%.

Benjamini-Hochberg procedure

Definition:

$$\text{False Discovery Rate (FDR)} = \mathbb{E}\left(\frac{V}{V+S}\right),$$

where V is false positive discoveries and S is the true positive discoveries.

FDR controls the **expectation proportion** of false discoveries among all reported discoveries.

Benjamini-Hochberg procedure

Suppose we reject all SNPs with $p_g \leq t$.

Under the null hypothesis, $p_g \sim \text{Uniform}(0, 1)$. So among G null SNPs, the expected number of false positives is approximately Gt.

If this threshold gives k discoveries, then the estimated false discovery proportion is roughly Gt/k.

To control this at level q, we want

$$\frac{Gt}{k} \leq q, \quad \text{which means} \quad t \leq \frac{k}{G} q.$$

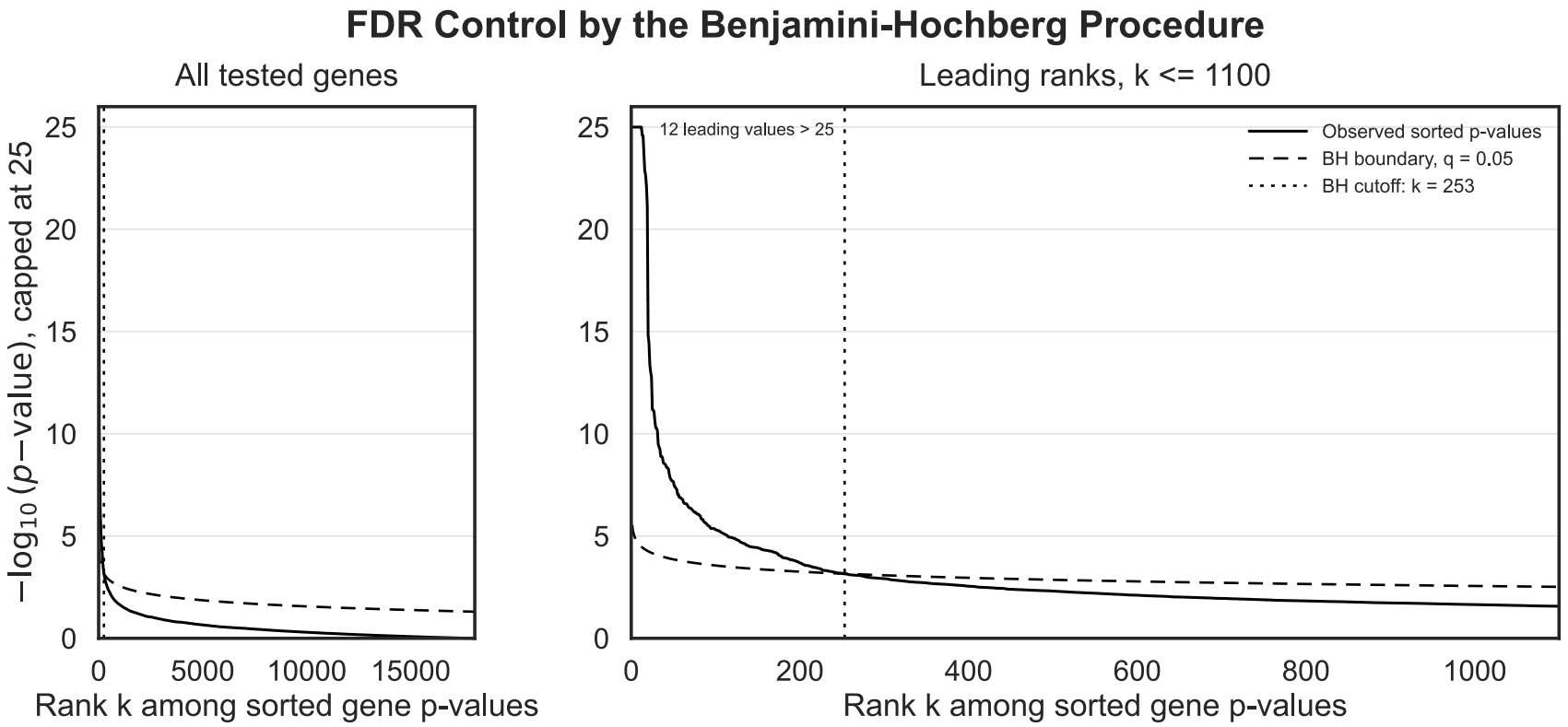
Benjamini-Hochberg procedure

Step 1: sort all p-values: $p_{(1)} \leq p_{(2)} \leq \ldots \leq p_{(G)}$.

Step 2: find the largest k such that $p_{(k)} \leq \frac{k}{G} q$.

Step 3: reject all hypotheses with $p_g \leq p_{(k)}$.

Plot: sorted p-values vs BH line



Raw p-value < 0.05: 1612
q-value < 0.05: 253

If we use FWER, 0.05/18080=2.77e-6, 89 genes remain

Bonferroni is more stringent than BH

FWER vs FDR: When to Use Which?

- **FWER:** controls the probability of making **at least one false discovery**.
  More stringent; fewer discoveries; higher confidence.
- **FDR:** controls the expected **fraction of false discoveries among all discoveries**.
  Less stringent; more discoveries; useful for exploratory screening.

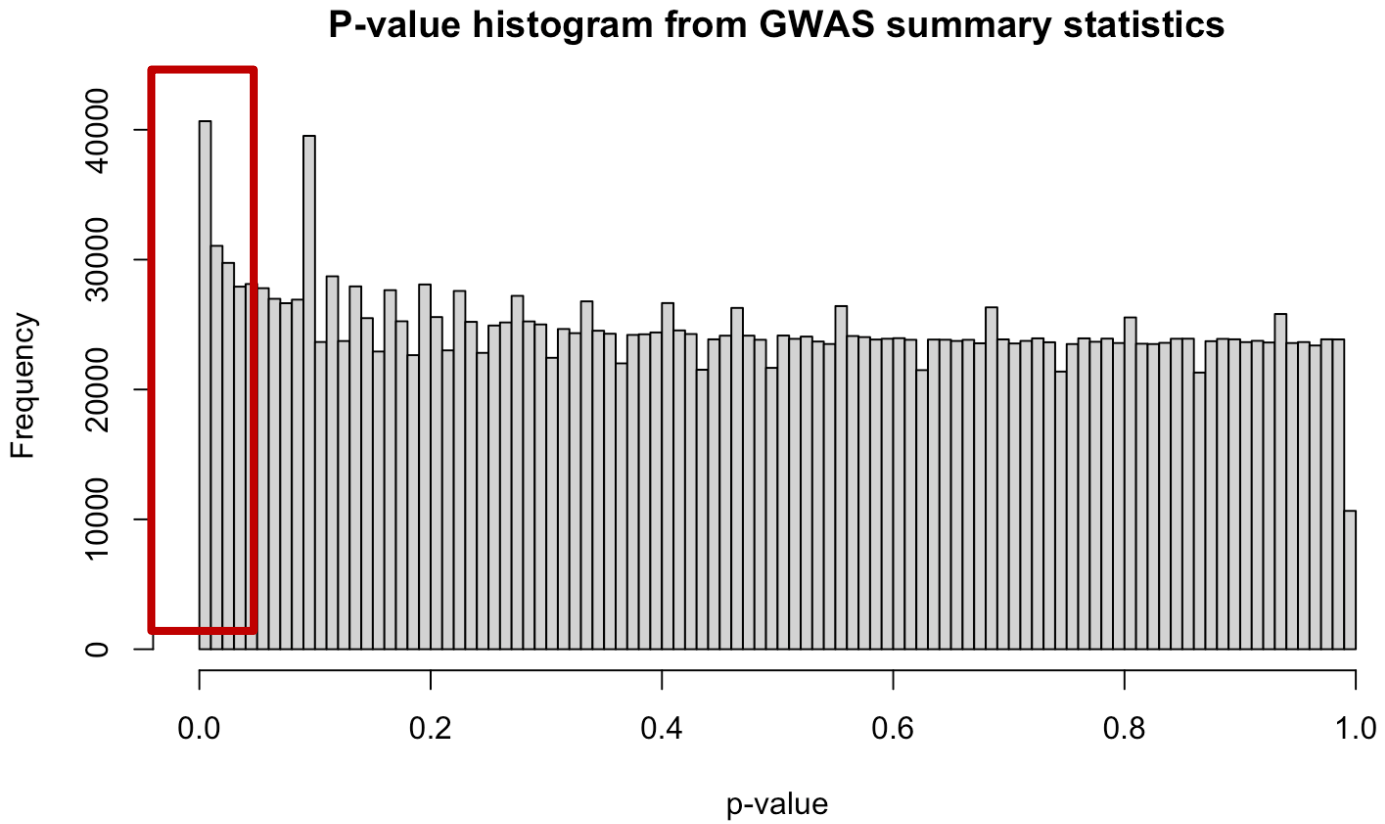
- **Use FWER** when a false positive is costly:
  genome-wide significant SNPs, clinical biomarkers, confirmatory studies.
- **Use FDR** when we want a candidate list:
  differential gene expression, single-cell marker genes, proteomics/metabolomics screening.

Comparing thresholds in this GWAS data

Method	**Criterion**	**Threshold**	**Number of SNPs**
Raw p-value	no correction	$p < 0.05$	157,514
Genome-wide	standard GWAS threshold	$p < 5 \times 10^{-8}$	236
Bonferroni	FWER	$p < 2.02 \times 10^{-8}$	222
B-H	FDR	$q < 0.05$	754

Recall: p-value histogram

In GWAS, we usually believe most SNPs are not associated; only a small fraction have nonzero effects, i.e., $\pi_i \approx 1$, where π_0 is the proportion of null hypotheses.



Storey intuition

In GWAS, we usually believe most SNPs are not associated; only a small fraction have nonzero effects, i.e., $\pi_i \approx 1$, where π_0 is the proportion of null hypotheses.

Under null hypotheses, p-values are from Uniform$(0, 1)$.

Storey idea: set $\lambda = 0.5$. Then

$$\hat{\pi}_0 = \frac{\#(p_g > \lambda)}{(1-\lambda)G}.$$

In our data, $\hat{\pi}_0 = 0.952$.

Main intuition: If only 95.2% of tests are null, we should not correct as if 100% are null.

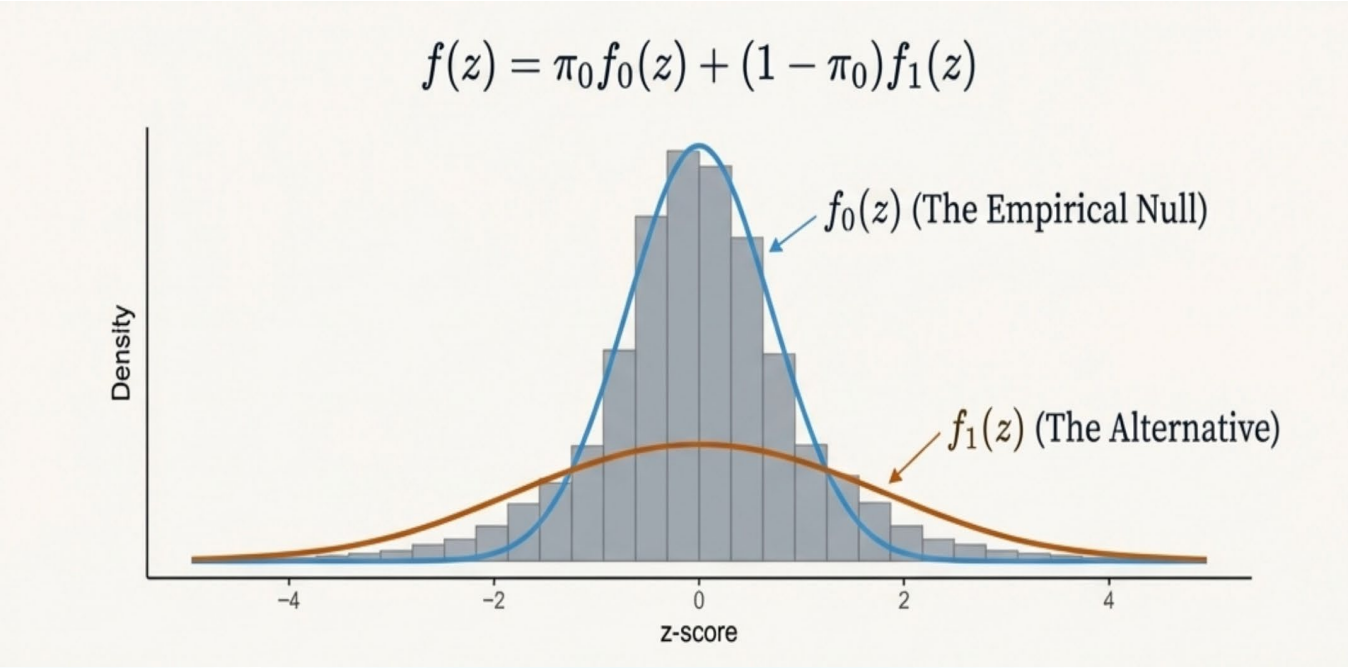
BH vs Storey

B-H threshold: $p_{(k)} \leq \frac{k}{m} q$.

Storey threshold: $p_{(k)} \leq \frac{k}{m} \frac{q}{\hat{\pi}_0}$.

Since $\hat{\pi}_0 < 1$, Storey's threshold is less conservative.

Empirical Bayes: Learning the Null from Your Data



Estimate Null distribution

Efron's two-groups model: mixture of null and non-null components.

Implicit assumption: the majority is noise

Local FDR

$$lfdr = P(H_0 | z)$$

R package: *locfdr*

Empirical Bayes: two-groups model

Empirical Bayes assumes that each SNP comes from one of two groups: null SNPs or signal SNPs.

The observed distribution of z-score is a mixture: $f(z) = \pi_0 f_0(z) + (1 - \pi_0) f_1(z)$, where $f_0(z)$ is the null distribution and $f_1(z)$ is the signal distribution.

For one SNP with test statistic z, empirical Bayes asks:

How likely is this SNP to be null?

That is the local false discovery rate

$$\text{lfdr}(z) = \Pr(H_0 \mid Z = z).$$

Empirical Bayes: two-groups model

The local false discovery rate

$$\mathrm{lfdr}(z) = \Pr(H_0 \mid Z = z) = \frac{\Pr(H_0) f_0(z)}{\Pr(H_0) f_0(z) + \Pr(H_1) f_1(z)} = \frac{\pi_0 f_0(z)}{f(z)}.$$

Reject the null hypothesis when lfdr is smaller than a threshold. For example, $\mathrm{lfdr}(z) \leq 0.1$ means this SNP has a false discovery probability less than 10%.

Comparsion with B-H: B-H controls the average false local discovery rate; local FDR gives a probability of each SNP.